

1. Ca up / PO4 down / PTH up banner

WHAT YOU SEE

Top banner: CA UP / PO4 DOWN / PTH UP-OR-NORMAL — the office's headline triad that defines primary hyperparathyroidism at a glance.

WHAT IT ENCODES

The defining biochemical triad of primary hyperparathyroidism: HIGH calcium, LOW phosphate (PTH drives proximal-tubule phosphate wasting), and HIGH or inappropriately-normal PTH. Chloride tends to be high and bicarbonate low (PTH causes a mild hyperchloremic metabolic acidosis; chloride:phosphate ratio >33 is a classic clue). Most cases are caught incidentally on routine chemistry in postmenopausal women.

BOARD TRAP

WRONG to dismiss a mid-normal PTH with hypercalcemia as 'no parathyroid problem.' Normal feedback should suppress PTH to near-zero when calcium is high, so a non-suppressed PTH IS the disease. Interpret PTH relative to calcium, not against the reference range.

2. Brown tumors

WHAT YOU SEE

Top-left BROWN TUMORS label over a brown lytic lesion in bone tied to OSTEITIS FIBROSA CYSTICA — the brown cavity in the bone is PTH-driven resorption replacing bone with fibrous, hemosiderin-stained tissue.

WHAT IT ENCODES

Brown tumors are localized lytic lesions of osteitis fibrosa cystica: PTH-driven osteoclastic resorption replaces bone with fibrovascular tissue and hemosiderin-laden giant cells, giving a brown gross appearance. They are a feature of SEVERE, advanced hyperparathyroidism (primary or, classically, renal/secondary-tertiary) and can fracture or be mistaken for primary bone tumors.

BOARD TRAP

WRONG to biopsy a brown tumor and call it a giant cell tumor or metastasis — they are histologically similar but the clue is the patient's high Ca and very high PTH. Check parathyroid biochemistry before resecting a presumed bone neoplasm; brown tumors regress after the hyperparathyroidism is treated.

3. Osteitis fibrosa cystica column

WHAT YOU SEE

A cracked, crumbling stone PILLAR labelled OSTEITIS FIBROSA CYSTICA with a salt-and-pepper skull — the failing column is end-stage high-turnover bone disease eaten away by chronic PTH.

WHAT IT ENCODES

Osteitis fibrosa cystica is the end-stage skeletal disease of hyperparathyroidism. Radiographic hallmarks: SUBPERIOSTEAL bone resorption along the radial side of the middle phalanges (most specific sign), 'salt-and-pepper' granular skull, distal clavicle tapering, and bone cysts/brown tumors. Reflects sustained high-turnover PTH effect.

BOARD TRAP

WRONG to read 'salt-and-pepper skull' as metastatic disease or multiple myeloma's discrete 'punched-out raindrop' skull lesions. Salt-and-pepper is a diffuse granular trabecular pattern specific to hyperparathyroidism; subperiosteal phalangeal resorption is the most specific radiographic sign — the resorption PATTERN discriminates.

4. Bone resorption worker

WHAT YOU SEE

A worker swinging a HAMMER to break apart a bone, labelled BONE RESORPTION — the demolition man is PTH-driven osteoclastic resorption liberating calcium from the skeleton.

WHAT IT ENCODES

PTH acts on osteoBLASTS (which carry the PTH receptor) to upregulate RANKL, which then activates osteoCLASTS to resorb bone and liberate calcium and phosphate. Continuous PTH excess is net CATABOLIC (cortical bone loss, especially distal radius), whereas INTERMITTENT PTH (teriparatide dosing) is paradoxically anabolic — the same hormone, opposite skeletal effect by exposure pattern.

BOARD TRAP

WRONG to assume osteoclasts have PTH receptors and are directly stimulated by PTH — they are activated INDIRECTLY via osteoblast-derived RANKL. This RANKL link is why denosumab (anti-RANKL) blunts the bone loss. The osteoblast-mediated mechanism is a favorite physiology distractor.

5. FHH familial gland / red X

WHAT YOU SEE

A parathyroid figure tagged FHH / FAMILIAL with 'urine Ca low, Ca/Cr <0.01' and a big red X over 'DON'T OPERATE' — the crossed-out scalpel warns that this body-wide CaSR defect is not cured by surgery.

WHAT IT ENCODES

Familial hypocalciuric hypercalcemia (autosomal dominant inactivating CaSR mutation) biochemically mimics mild primary HPT (high Ca, high/normal PTH) but has LOW urine calcium (24-h Ca/Cr clearance ratio <0.01) and is benign/lifelong. Surgery is NOT indicated and does not work.

BOARD TRAP

WRONG to parathyroidectomize an FHH patient — because the CaSR defect is in every tissue (parathyroid AND kidney), removing glands leaves calcium high. Always rule out FHH with the urine Ca/Cr clearance ratio (and family history) before referring for surgery: <0.01 = FHH (do NOT operate), >0.02 = primary HPT.

6. Classic tetrad: stones/bones/groans/psych

WHAT YOU SEE

Central list on the wall: STONES, BONES, abdominal GROANS, psychiatric overtones — the four-part roster of symptomatic hypercalcemia hung in the office.

WHAT IT ENCODES

Symptomatic hypercalcemia of primary HPT: 'stones' (calcium nephrolithiasis, nephrocalcinosis, nephrogenic DI causing polyuria), 'bones' (osteitis fibrosa cystica, osteoporosis), 'abdominal groans' (constipation, anorexia, peptic ulcer via gastrin, acute pancreatitis), and 'psychiatric overtones' (fatigue, depression, confusion). Severe acute hypercalcemia also shortens the QT interval.

BOARD TRAP

WRONG to expect ECG QT PROLONGATION in hypercalcemia — calcium SHORTENS the QT (it lengthens with hypoCalcemia). Mixing up the QT direction is a classic trap; remember high Ca = short QT.

7. Adenoma boss

WHAT YOU SEE

A smug parathyroid 'boss' sitting at the executive desk labelled ADENOMA / AUTONOMOUS — the gland acting like an unaccountable CEO pictures autonomous PTH secretion that ignores calcium feedback.

WHAT IT ENCODES

A single autonomous parathyroid adenoma causes ~80–85% of primary hyperparathyroidism; four-gland hyperplasia ~10–15%, and parathyroid carcinoma <1% (suspect with very high Ca, very high PTH, and a palpable neck mass). The adenoma secretes PTH independent of feedback, so calcium rises while PTH stays inappropriately non-suppressed.

BOARD TRAP

WRONG to assume EVERY primary HPT is a sporadic single adenoma — multigland HYPERPLASIA points to a familial syndrome: MEN1 (parathyroid + pituitary + pancreas, the '3 Ps'), MEN2A (parathyroid + medullary thyroid CA + pheochromocytoma). In young/familial/multigland disease, screen for MEN and check the RET (MEN2A) or MEN1 gene.

8. Renal PO4 wasting / down PO4

WHAT YOU SEE

A floor DRAIN with PO4 swirling down it, marked RENAL PO4 WASTING / PO4 DOWN — the drain literally is PTH dumping phosphate into the urine, the partner of the rising calcium.

WHAT IT ENCODES

PTH inhibits the proximal-tubule sodium-phosphate cotransporter (NaPi-IIa), causing phosphaturia and hypophosphatemia. Simultaneously PTH increases DISTAL-tubule calcium reabsorption (raising serum Ca) and stimulates renal 1-alpha-hydroxylase to make calcitriol. So PTH lowers phosphate while raising calcium — the inverse Ca/PO4 movement is the renal signature.

BOARD TRAP

WRONG to expect high phosphate in primary HPT — that pattern (high Ca AND high PO4) suggests vitamin D toxicity or milk-alkali instead, where there is no PTH-driven phosphate wasting. Low phosphate is the expected partner of PTH-driven hypercalcemia.

9. Gut absorption / 1,25-D up

WHAT YOU SEE

A glowing oil LAMP labelled 1,25-D casting light onto the gut with an up-arrow — the lamp is PTH-stimulated calcitriol, PTH's INDIRECT way of boosting intestinal calcium absorption.

WHAT IT ENCODES

PTH stimulates renal 1-alpha-hydroxylase to convert 25-OH-D into active 1,25-(OH)2-D (calcitriol), which then increases INTESTINAL absorption of calcium and phosphate. PTH has no direct action on the gut — its intestinal effect is entirely INDIRECT through calcitriol. This is the third arm of PTH's calcium-raising action (bone, distal kidney, and gut-via-vitamin-D).

BOARD TRAP

WRONG to state PTH directly increases gut calcium absorption — it does so only by way of increased calcitriol. The PTH→1-alpha-hydroxylase→1,25-D→gut chain is the indirect pathway tested as a distractor.

10. Definitive: parathyroidectomy

WHAT YOU SEE

Top-right tag DEFINITIVE: PARATHYROIDECTOMY — marks surgery as the only curative option once any operative criterion is met.

WHAT IT ENCODES

Parathyroidectomy is the ONLY curative therapy. Surgery is indicated for symptomatic disease (stones, fractures) and, in asymptomatic patients, when ANY of: serum Ca >1 mg/dL above upper normal; T-score ≤ -2.5 or fragility fracture or vertebral fracture on imaging; eGFR <60, 24-h urine Ca >400 mg with stone risk, or nephrolithiasis/nephrocalcinosis; or age <50. Intraoperative PTH that falls >50% confirms cure.

BOARD TRAP

WRONG to tell a young (<50) asymptomatic patient with mild hypercalcemia that they just need monitoring — age under 50 is itself a surgical indication because of the long lifetime risk. Memorize the specific cutoffs; 'asymptomatic' does not automatically mean 'observe.'

11. Cinacalcet / CaSR

WHAT YOU SEE

A pill bottle labelled CINACALCET / CaSR (NON-SURGICAL) — the calcimimetic that sensitizes the calcium receptor to lower PTH and calcium without removing the gland.

WHAT IT ENCODES

Cinacalcet is a CALCIMIMETIC that allosterically sensitizes the calcium-sensing receptor to ambient calcium, so the parathyroids 'see' more calcium and lower PTH and serum Ca. Used when surgery is contraindicated/declined, in parathyroid carcinoma, and in secondary HPT of dialysis CKD. It lowers calcium but does NOT improve bone density.

BOARD TRAP

WRONG to expect cinacalcet to RAISE bone density or cure the disease — it controls calcium/PTH but leaves the adenoma in place and does not rebuild bone. For an operative candidate, surgery (curative, bone-protective) is preferred over lifelong cinacalcet. Confusing 'controls labs' with 'cures' is the trap.

12. Bisphosphonate

WHAT YOU SEE

A pill/IV bottle labelled BISPHOSPHONATE — the antiresorptive that protects bone density but does not cure the over-secreting adenoma.

WHAT IT ENCODES

Bisphosphonates (alendronate orally, zoledronic acid IV) inhibit osteoclastic resorption, protecting/improving bone density and acutely lowering malignant hypercalcemia. In primary HPT they are an adjunct for the skeleton in non-surgical patients but do not normalize PTH or cure the disease.

BOARD TRAP

WRONG to think bisphosphonates or cinacalcet 'treat' primary HPT definitively — only removing the autonomous gland cures it. Medical therapy targets symptoms (bone or calcium) while the underlying adenoma keeps over-secreting PTH. Symptom control ≠ cure.

13. Sestamibi / localize not diagnose

WHAT YOU SEE

A bottom monitor/camera on a tripod labelled SESTAMIBI with the caption 'LOCALIZE, NOT DIAGNOSE' — the imaging camera only finds the gland after biochemistry has already made the diagnosis.

WHAT IT ENCODES

Technetium-99m sestamibi scintigraphy (often with SPECT/CT) and neck ultrasound are LOCALIZATION studies done AFTER the biochemical diagnosis, to guide minimally invasive single-gland parathyroidectomy. Sestamibi is retained by mitochondria-rich adenoma cells. The diagnosis itself is made by serum Ca and PTH.

BOARD TRAP

WRONG to use a negative sestamibi scan to EXCLUDE primary HPT — imaging neither confirms nor refutes the diagnosis, which is purely biochemical (high Ca + non-suppressed PTH). A negative scan just means a four-gland bilateral neck exploration may be needed, not that the patient lacks disease.

14. Hungry bone post-op

WHAT YOU SEE

A patient asleep in bed dreaming of Ca DOWN / PO4 DOWN, labelled HUNGRY BONE — the dreaming figure is demineralized bone gobbling up calcium and phosphate after the PTH drive is suddenly removed.

WHAT IT ENCODES

Hungry bone syndrome is profound, PROLONGED postoperative hypocalcemia (with hypophosphatemia and hypomagnesemia) after parathyroidectomy for long-standing hyperparathyroidism: once the PTH drive is removed, demineralized bone avidly re-uptakes calcium and phosphate. Highest risk with high pre-op PTH/ALP and osteitis fibrosa cystica. Treat with aggressive calcium and calcitriol, and replete magnesium.

BOARD TRAP

WRONG to attribute persistent post-op hypocalcemia solely to surgical hypoparathyroidism (accidental removal/devascularization of normal glands) without checking PTH — in hypoPARAthyroidism PTH is LOW, whereas in hungry bone PTH is normal/high with low Ca AND low PO4. The PTH level and phosphate direction distinguish the two.